EE5502
MOS Devices

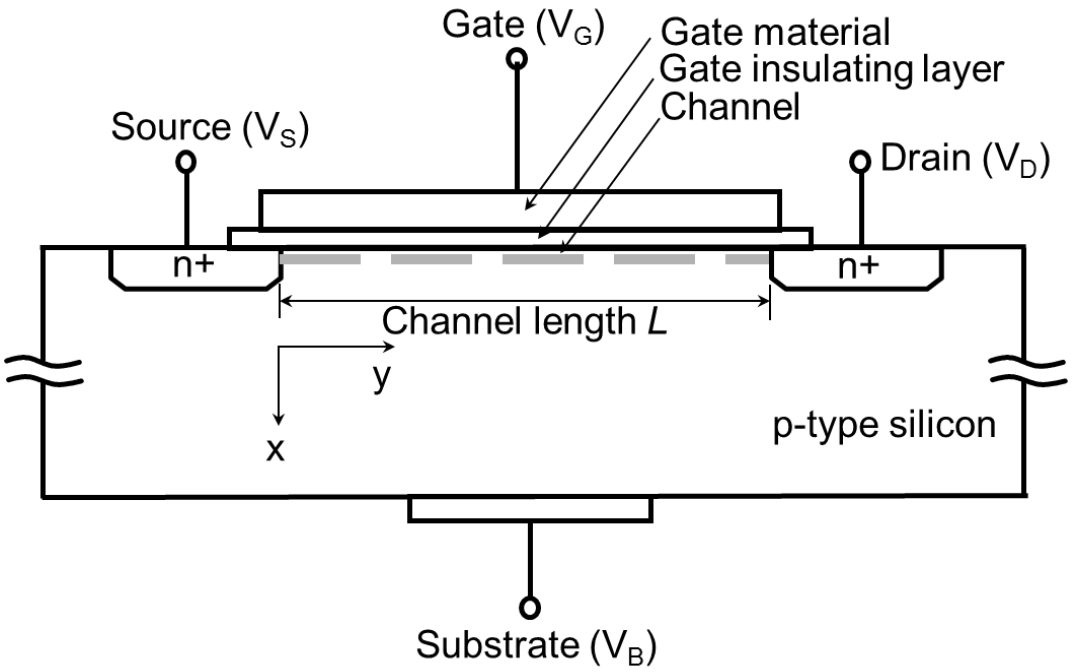
Long Channel MOSFETs
- Basic MOSFET Operation

Zhu Chunxiang
Department of Electrical and Computer Engineering
National University of Singapore
Email: elezhucx@nus.edu.sg
Phone: 6516 8930
Office: E1-08-08

Long Channel MOSFETs

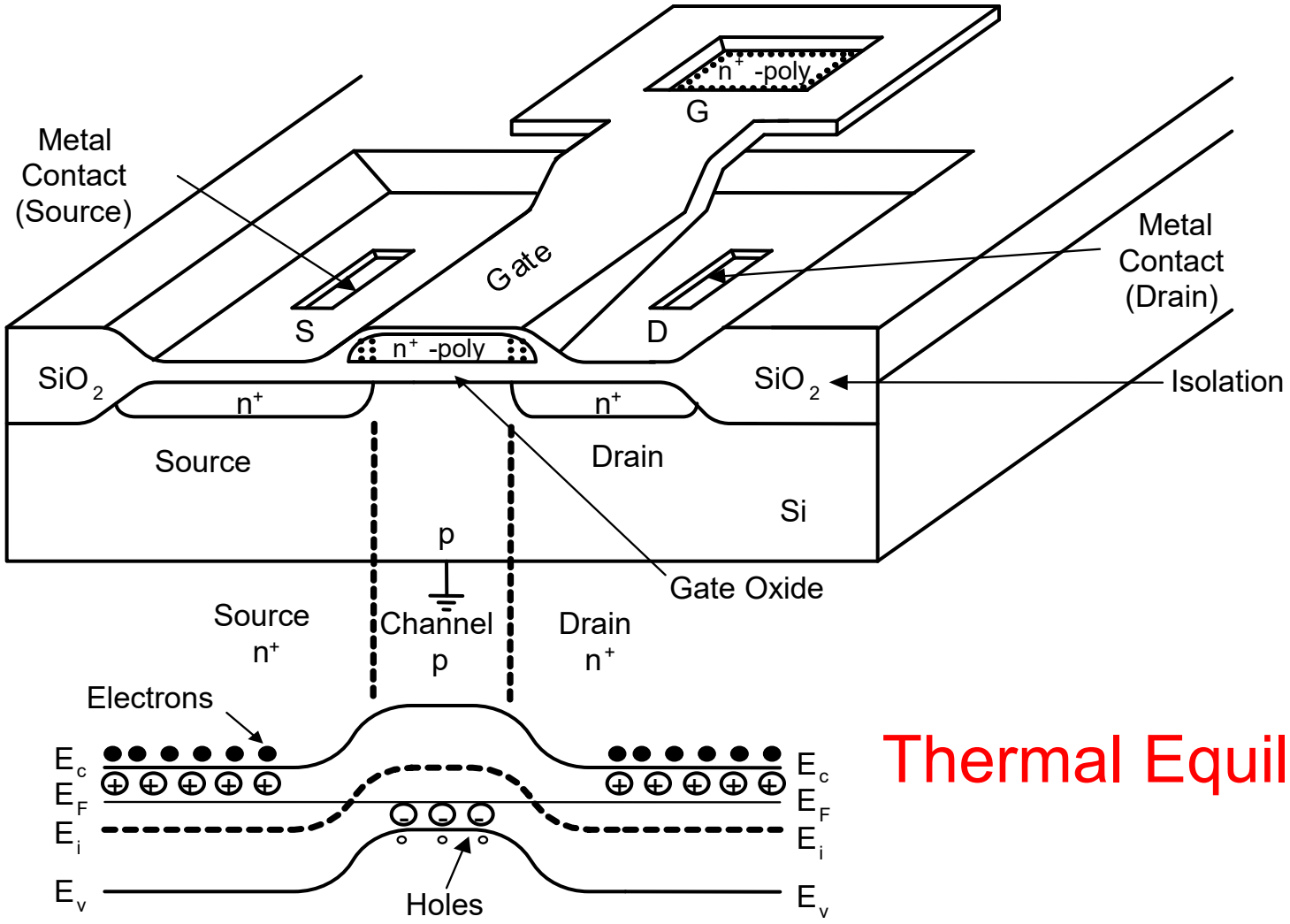
- **Basic MOSFET Operation**
- Drain Current Models
- Channel Length Modulation
- Threshold Voltage
- Sub-threshold Characteristics
- Body (Substrate Bias) Effect
- MOSFET Channel Mobility

MOSFET Structure



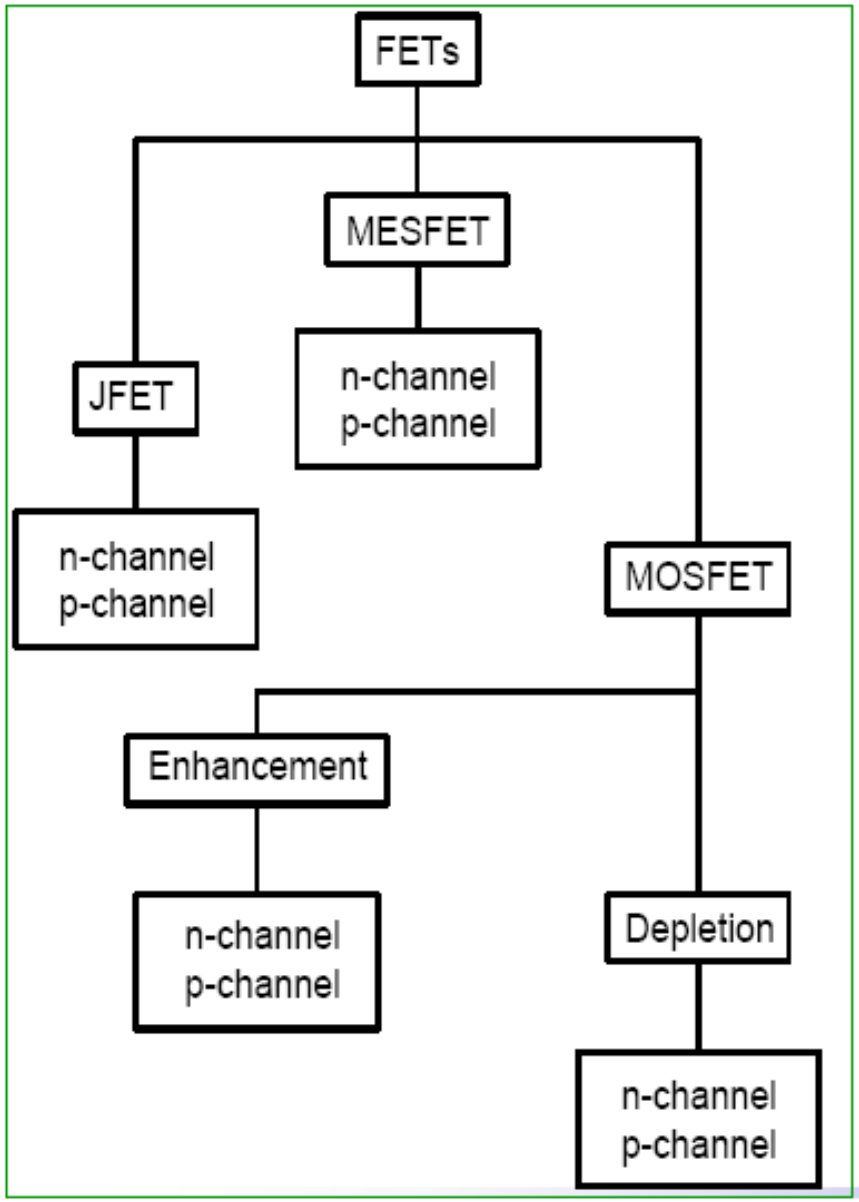
- Gate voltage is used to induce inversion electrons (NMOS) and holes (PMOS) by capacitive coupling
- A horizontal electric field (V_{DS}) then drives the inversion carriers to flow from source to drain by drift current
- MOSFET operates in single carrier configuration: unipolar device (electrons for n-channel MOSFET and holes for p-channel)

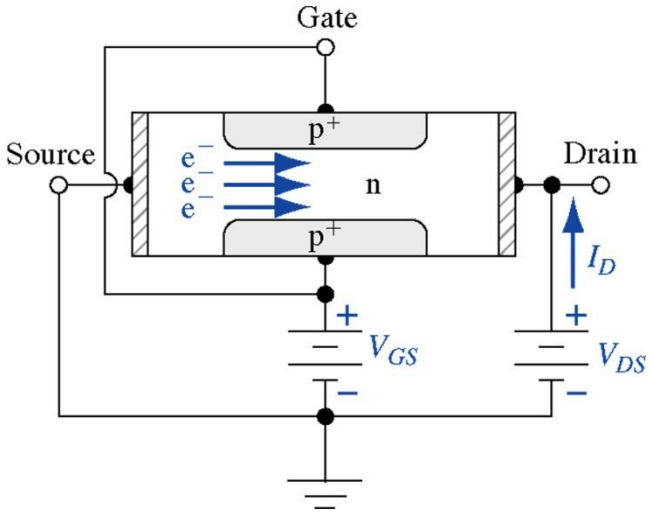
3D MOSFET Structure



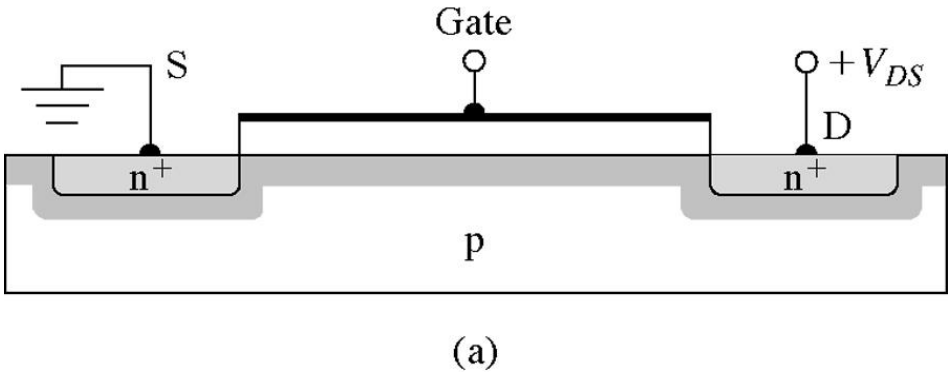
Thermal Equilibrium

Family of FETs



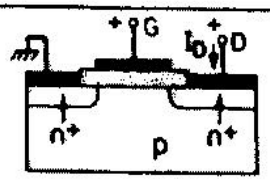
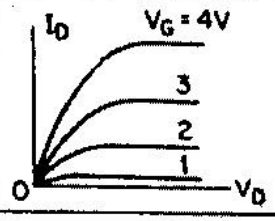
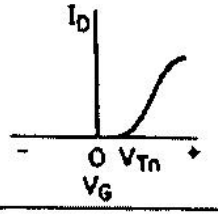
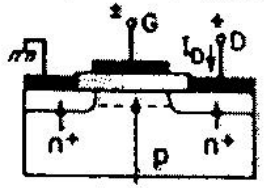
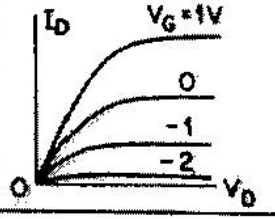
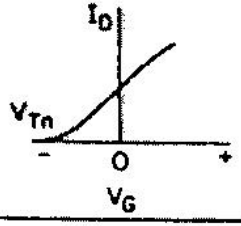
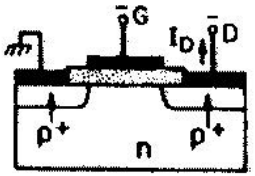
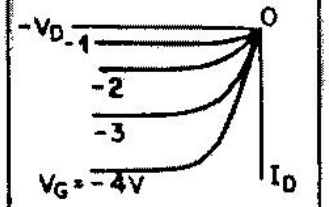
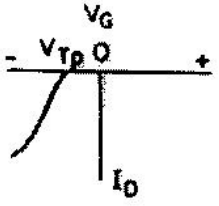
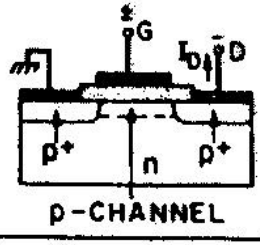
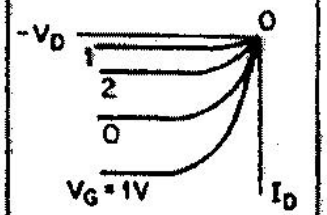
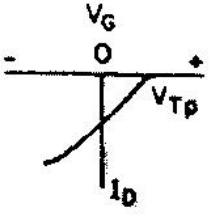


JFET (Junction FET)

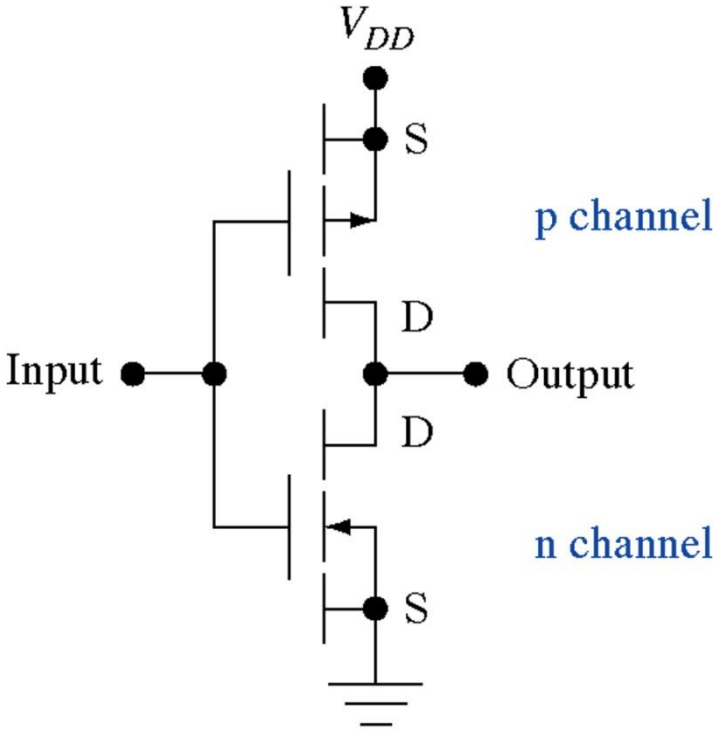


MOSFET (Metal Oxide Semiconductor FET)

Classification of MOSFET

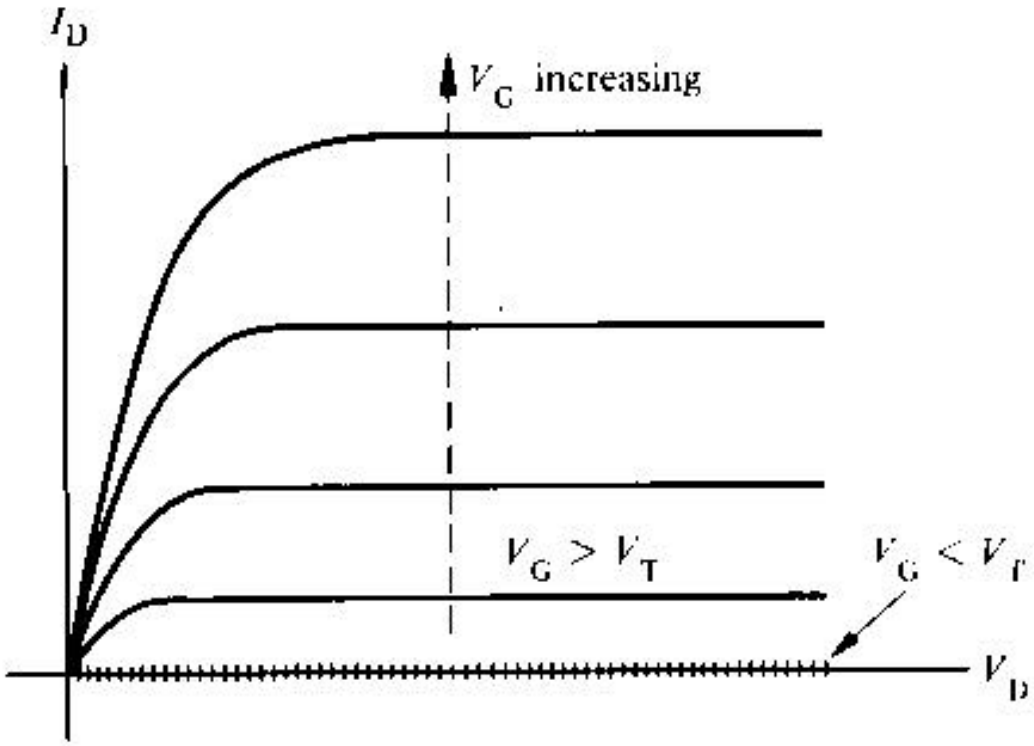
TYPE	CROSS SECTION	OUTPUT CHARACTERISTICS	TRANSFER CHARACTERISTICS
n-CHANNEL ENHANCEMENT (NORMALLY OFF)			
n-CHANNEL DEPLETION (NORMALLY ON)			
p-CHANNEL ENHANCEMENT (NORMALLY OFF)			
p-CHANNEL DEPLETION (NORMALLY ON)			

Basic MOSFET Characteristics



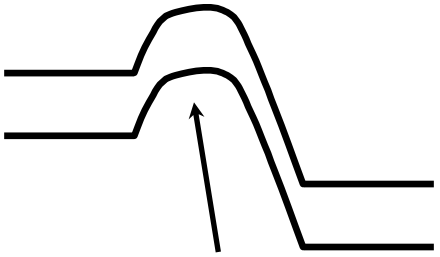
Circuit symbol of n & p channel enhancement mode MOSFET

Output Current-Voltage characteristics



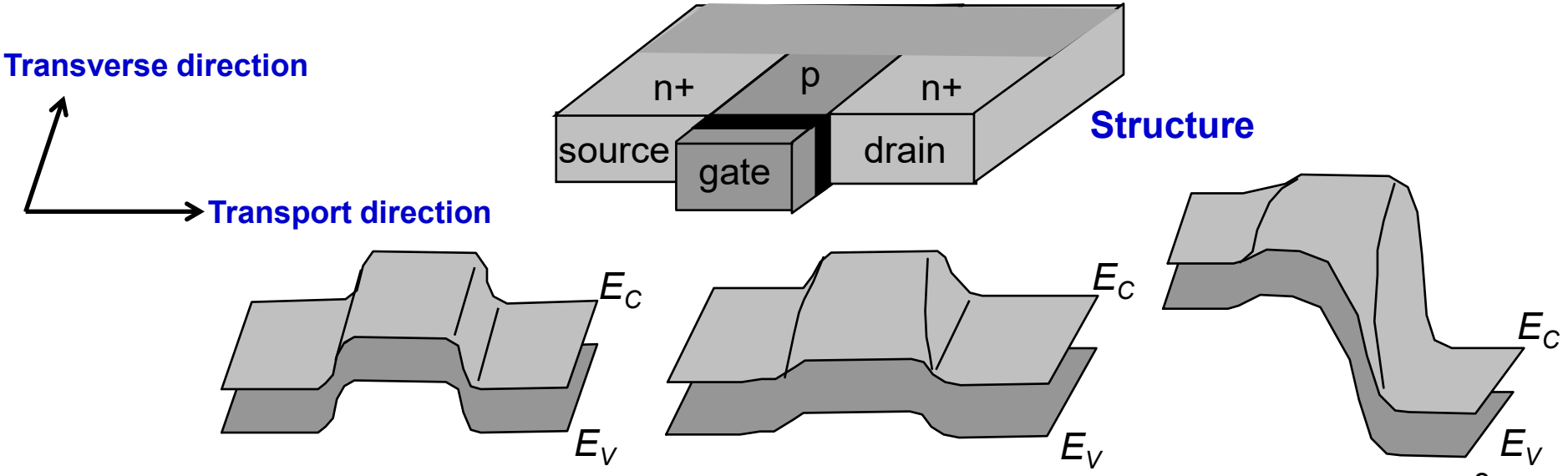
Basic MOSFET Operation

- **Basic idea**



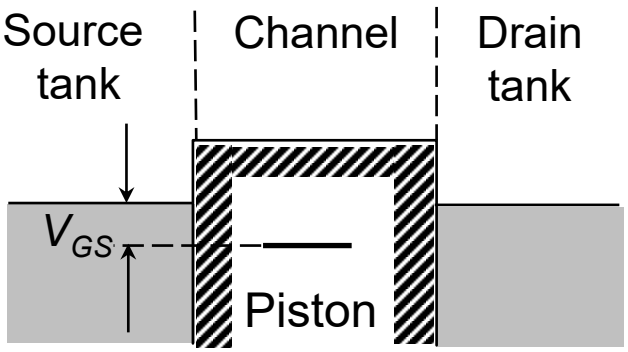
Barrier controlled by
3rd terminal (gate)

- The gate voltage is “increased” to induce carriers (electrons) by capacitive coupling (i.e., the potential barrier is lowered).
- Electrons injected from the source will be collected at the drain.

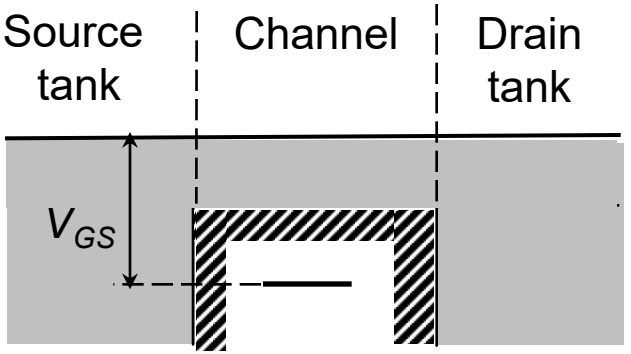


Basic MOSFET Operation

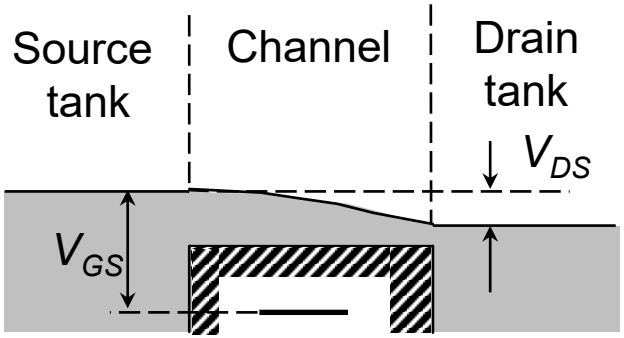
- **Water analogy**



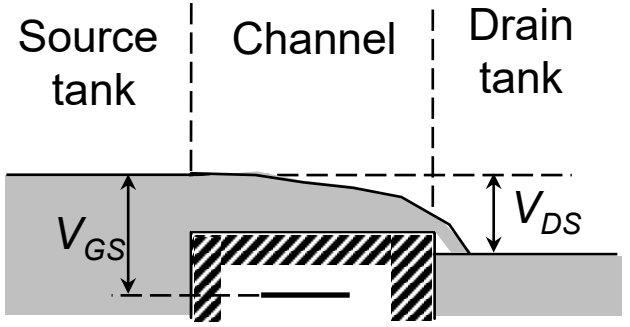
(a) Cutoff



(b) With channel strongly inverted, but no current since V_{DS}=0

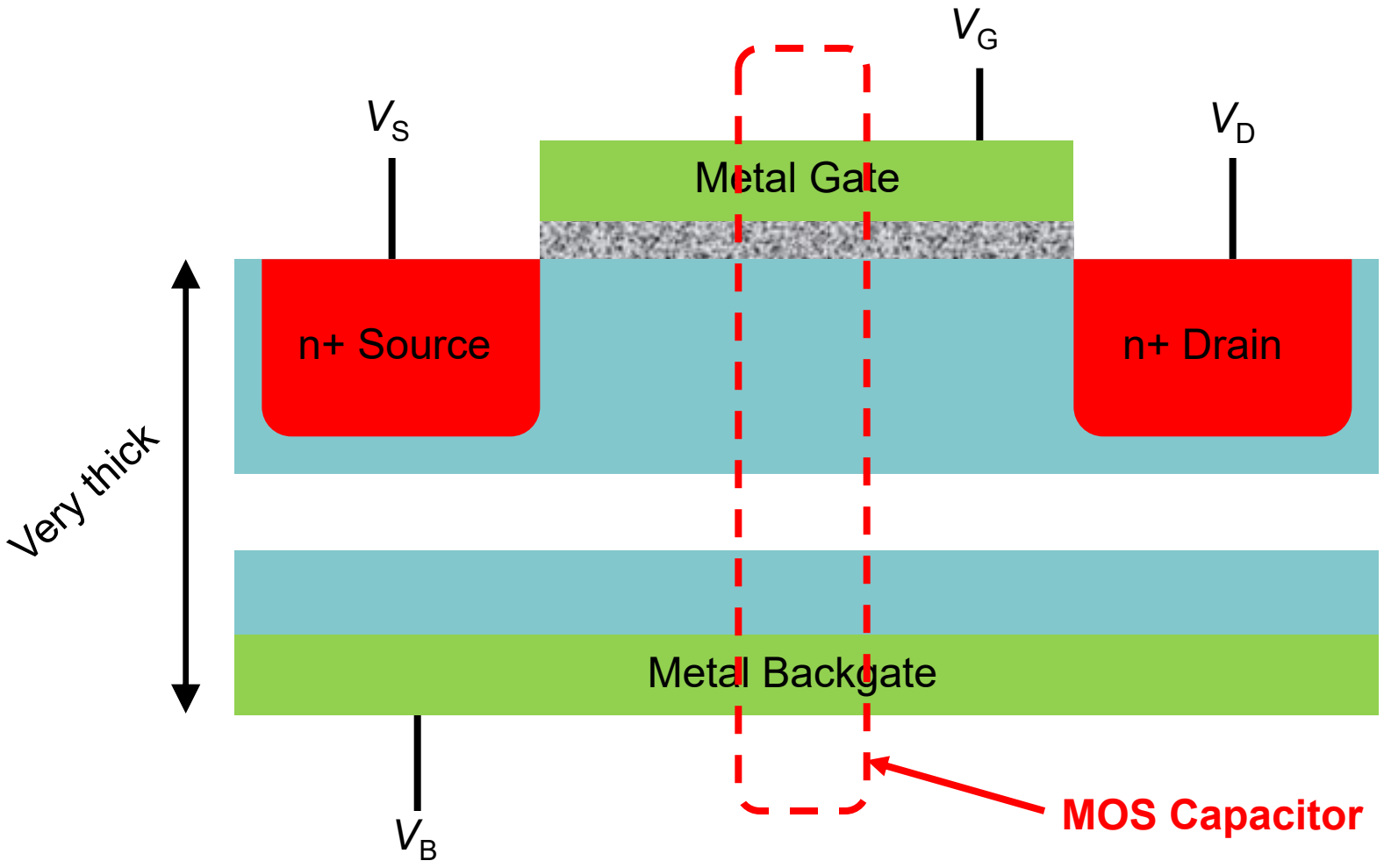


(c) Linear region with small V_{DS}



(d) Saturation

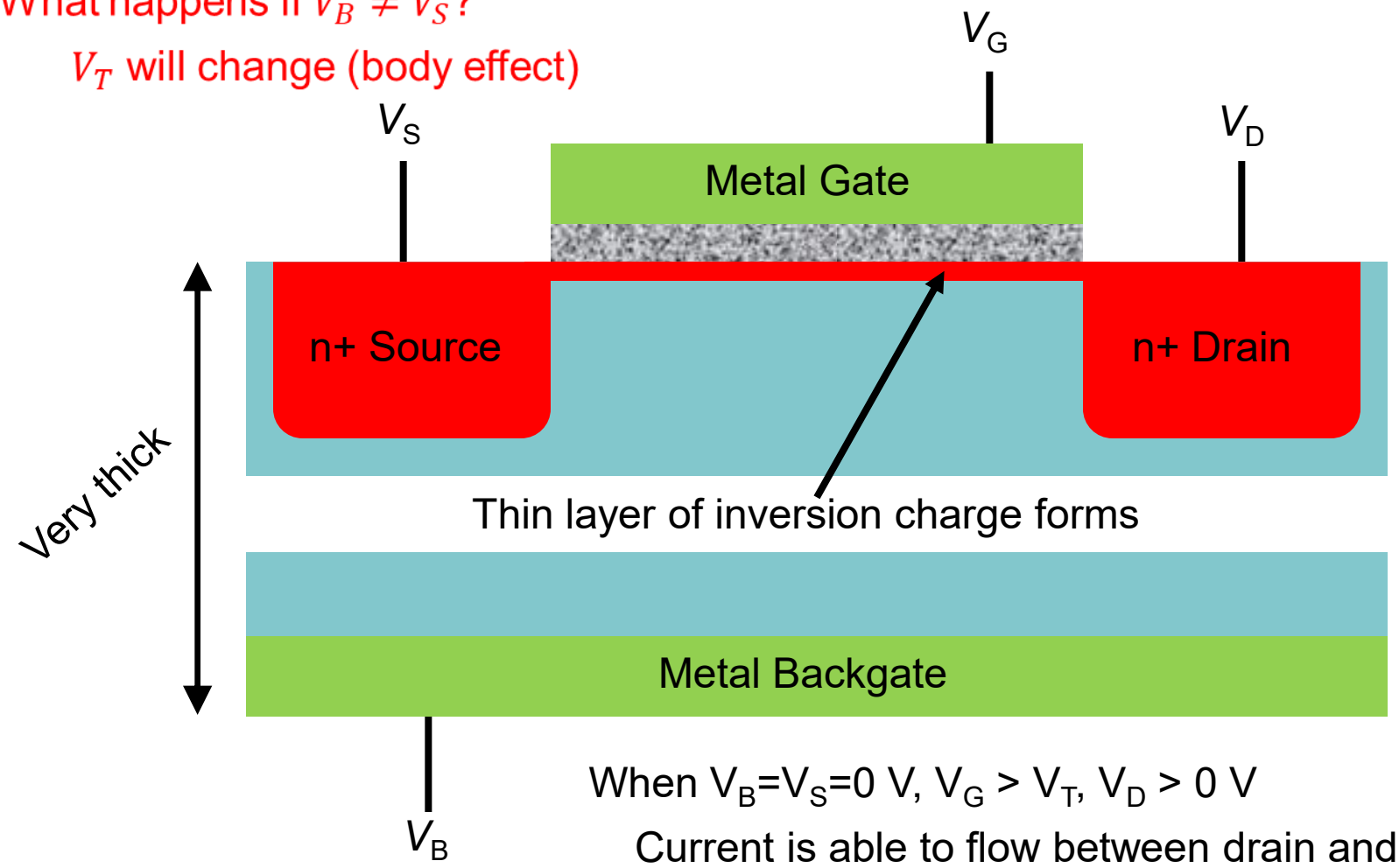
Basic MOSFET Operation – Simplified



Basic MOSFET Operation - Simplified

What happens if $V_B \neq V_S$?

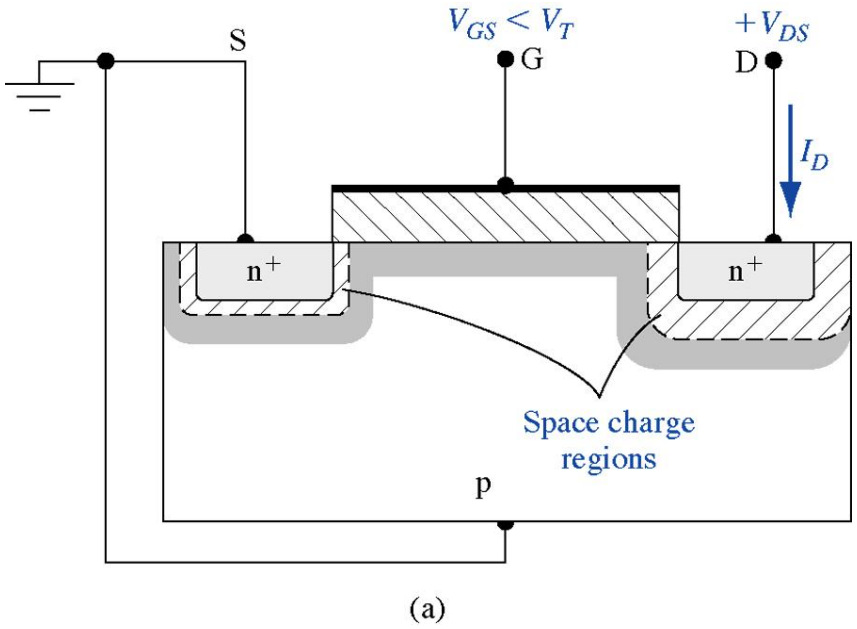
V_T will change (body effect)

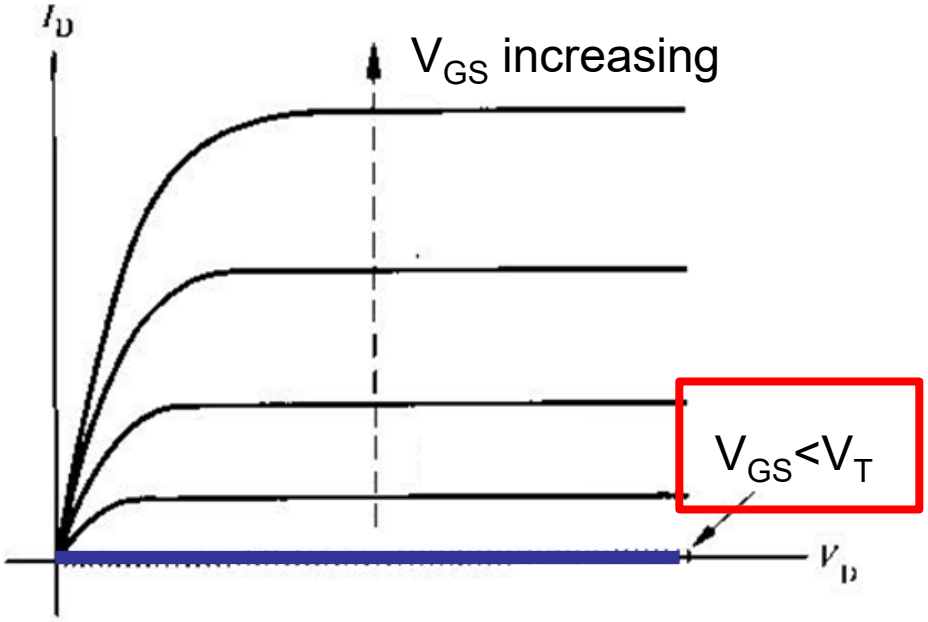


When V_B=V_S=0 V, V_G > V_T, V_D > 0 V

Current is able to flow between drain and source.

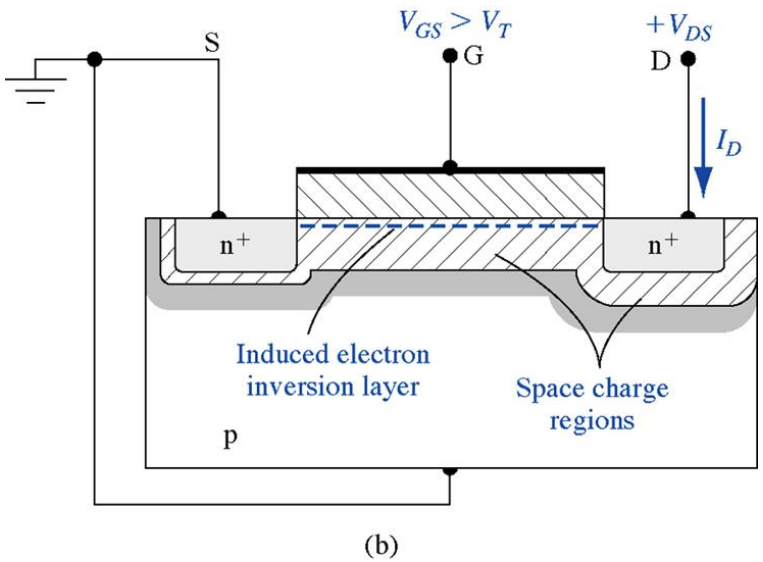
Cut-off region ($V_{GS} < V_T$)





- Source and Substrate terminals are held at ground potential
- No electron inversion layer since $V_{GS}<V_T$
- PN junction at drain is reverse biased
- Zero drain current, ignoring P-N junction leakage current

Linear region ($V_{GS} > V_T$, $V_{DS} < V_{DS(sat)}$)



$$I_D = g_d \cdot V_{DS}$$

g_d : Channel Conductance

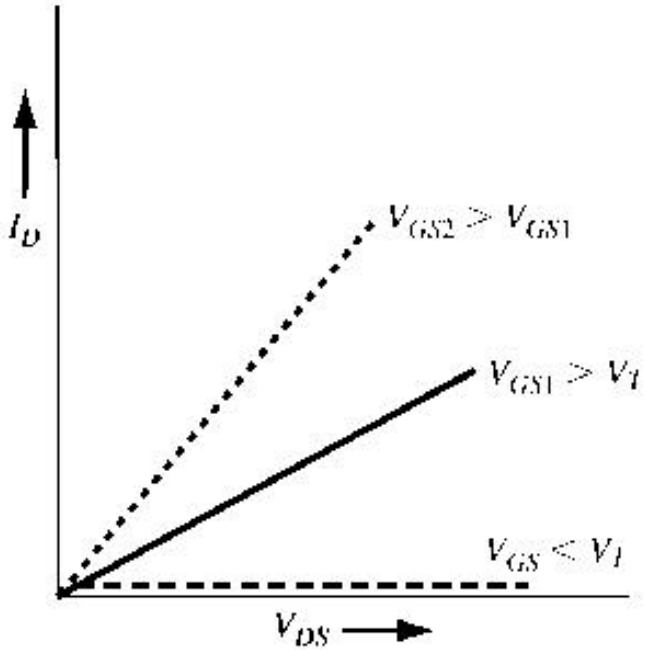
$$g_d = \frac{W}{L} \cdot \mu_n \cdot |Q_n|$$

μ_n : Mobility of electron in inversion layer

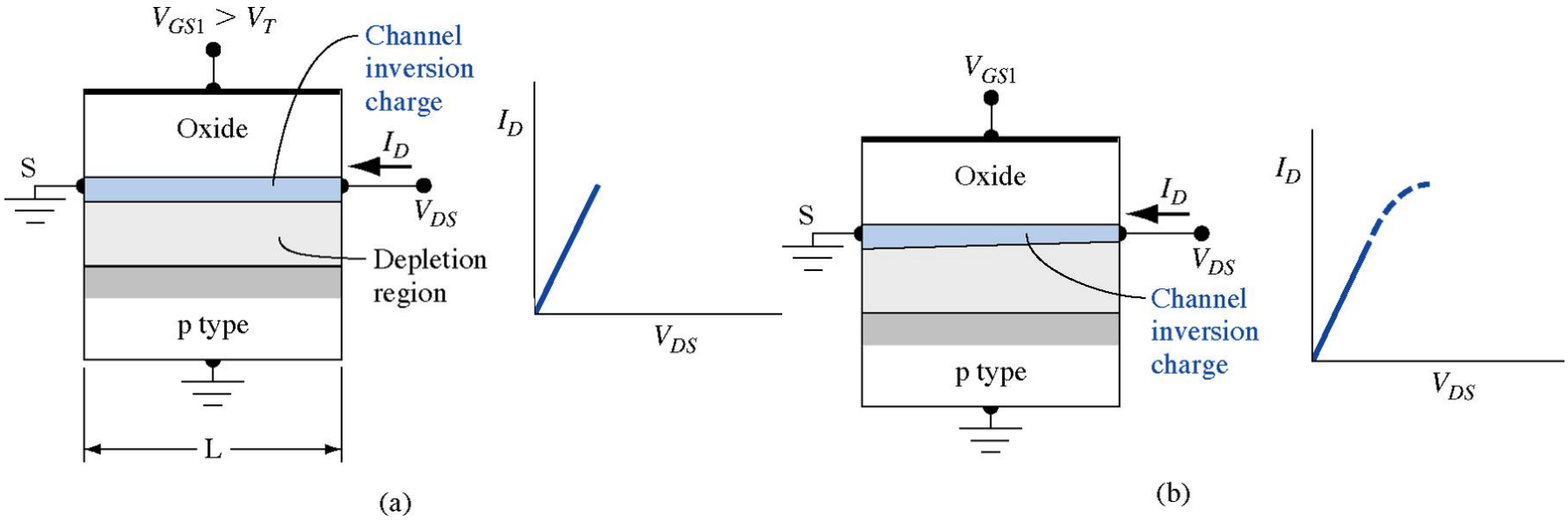
$|Q_n|$: Magnitude of inversion layer charge per area

$$|Q_n| = f(V_{GS}) \rightarrow g_d = f(V_{GS}) \rightarrow I_D = f(V_{GS})$$

- Electron inversion layer created.
- Electron will flow from Source (S) to the Drain (D) with small V_{DS}.
- Thickness of the inversion channel layer is constant along the channel length.
- Channel region has resistor characteristics.
- Basic MOSFET action: modulation of channel conductance by V_{GS}.

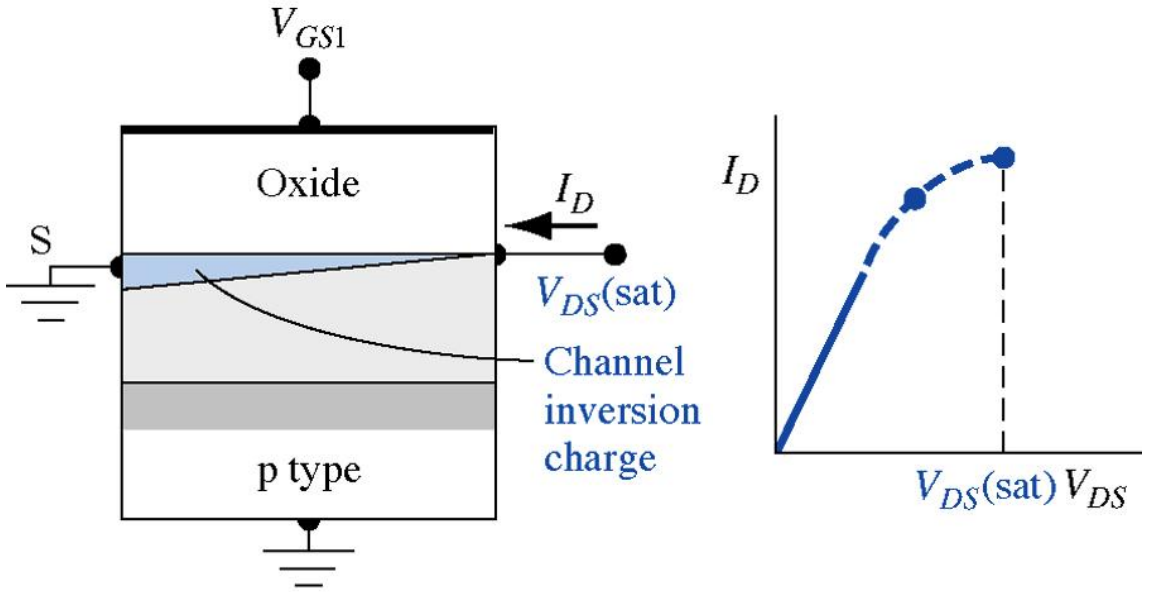


- As V_{DS} increases, the potential at Si surface near the drain increases, so the voltage drop across oxide at drain end decrease, resulting in the decrease in induced inversion charge at Si surface near drain end.
- Thus, incremental conductance of channel (slope of I_D – V_{DS} curve) decrease (i.e. slope reduces)



Cross section and I_D – V_{DS} curves for (a) small V_{DS} and (b) larger V_{DS}

Pinch-off voltage ($V_{GS} > V_T$, $V_{DS} = V_{DS(sat)}$)



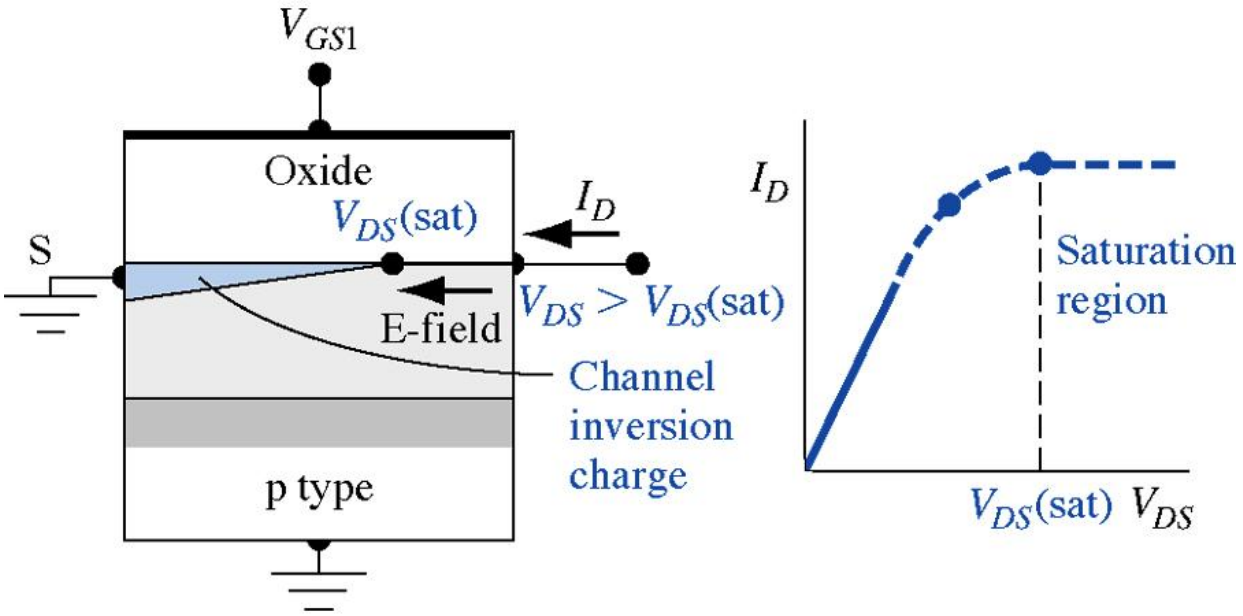
(c)

Cross section and $I_D - V_{DS}$ curve for (c) $V_{DS} = V_{Ds(Sat)}$

- Inversion charge at drain terminal becomes zero when voltage drop across oxide at drain terminal becomes V_T, as V_{DS} increase further to a certain voltage called $V_{DS(sat)}$.
- $V_{DS(sat)}$ can be defined as: $V_{DS(sat)} = V_{GS} - V_T$
- Conducting channel (inversion region) disappears at the drain end.

Saturation region ($V_{GS} > V_T$, $V_{DS} > V_{DS(sat)}$)

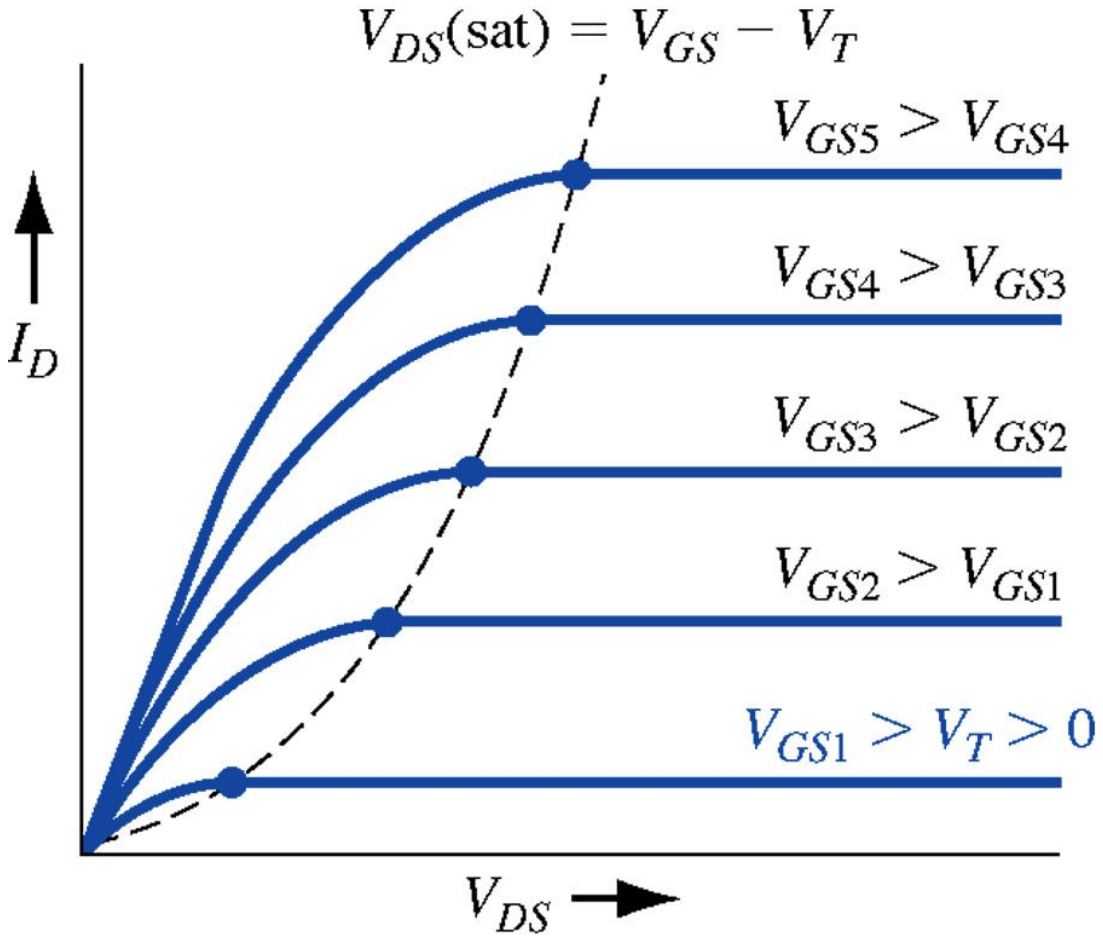
- For V_{DS} in excess of pinch-off voltage, $V_{DS(sat)}$, the pinched-off region widens into depleted channel (ΔL).
- ΔL section absorbs most of the voltage drop in excess of $V_{DS(sat)}$, the electrons are injected and swept by high E-field to drain contact
- Assuming $\Delta L << L$, I_D will be constant.



(d)

Cross section and $I_D - V_{DS}$ curve for (d) $V_{DS} > V_{Ds(Sat)}$

I_D-V_{DS} Characteristics



- I_D modulated by input voltage applied to gate, which generate different I_D – V_{DS} curves